innately specified; as, indeed, vertical faculty theorists from Gall to Chomsky have been wont to claim.

I have been arguing that the psychological systems whose operations "present the world to thought" constitute a natural kind by criteria independent of their similarity of function; there appears to be a cluster of properties that they have in common but which, *qua* input analyzers, they might perfectly well not have shared.[35] We can abbreviate all this by the claim that the input systems constitute a family of modules: domain-specific computational systems characterized by informational encapsulation, high-speed, restricted access, neural specificity, and the rest.

Let's suppose, probably contrary to fact, that you have found this story convincing. So, you are pretending to believe, for purposes of the following discussion, that the input systems are modular. If you actually did believe this, you would surely be led to pose the following question: are cognitive mechanisms *other than* input systems also modular? Or are the properties of being modular and being an input system coextensive? We are thus, finally, about to raise what I regard as the main issue: whether modularity is (as Gall, for example, thought it was) the *general* fact about the organization of the mind. I am going to suggest that at least some cognitive systems are nonmodular, and then I'm going to explore a variety of consequences of their (putative) nonmodularity.

PART IV
CENTRAL SYSTEMS

Vertical faculties are domain specific (by definition) and modular (by hypothesis). So the questions we now want to ask can be put like this: Are there psychological processes that can plausibly be assumed to cut across cognitive domains? And, if there are, is there reason to suppose that such processes are subserved by nonmodular (e.g., informationally unencapsulated) mechanisms?

The answer to the first of these questions is, I suppose, reasonably clear. Even if input systems are domain specific, there must be some cognitive mechanisms that are not. The general form of the argument goes back at least to Aristotle: the representations that

input systems deliver have to interface somewhere, and the computational mechanisms that effect the interface must ipso facto have access to information from more than one cognitive domain. Consider:

(a) We have repeatedly distinguished between what the input systems compute and what the organism (consciously or subdoxastically) *believes*. Part of the point of this distinction is that input systems, being informationally encapsulated, typically compute representations of the distal layout on the basis of less information about the distal layout than the organism has available. Such representations want correction in light of background knowledge (e.g., information in memory) and of the simultaneous results of input analysis in other domains (see Aristotle on the 'common sense'). Call the process of arriving at such corrected representations "the fixation of perceptual belief." To a first approximation, we can assume that the mechanisms that effect this process work like this: they look simultaneously at the representations delivered by the various input systems and at the information currently in memory, and they arrive at a best (i.e., best available) hypothesis about how the world must be, given these various sorts of data.[36] But if there are mechanisms that fix perceptual belief, and if they work in anything like this way, then these mechanisms are not domain specific. Indeed, the point of having them is precisely to ensure that, wherever possible, what the organism believes is determined by all the information it has access to, regardless of which cognitive domains this information is drawn from.

(b) We use language (inter alia) to communicate our views on how the world is. But this use of language is possible only if the mechanisms that mediate the production of speech have access to what we see (or hear, or remember, or think) that the world is like. Since, by assumption, such mechanisms effect an interface among vertical faculties, they cannot themselves be domain specific. More precisely, they must at least be *less* domain specific than the vertical faculties are.[37]

(c) One aspect of the 'impenetrability' of the input systems is, we assumed, their insensitivity to the utilities of the organism. This assumption was required in part to explain the *veridicality* of perception given that the world doesn't always prove to be the way that we would prefer it to be. However, an interface between per-

ception and utilities must take place *somewhere* if we are to use the information that input systems deliver in order to determine how we ought to act. (Decision theories are, to all intents and purposes, models of the structure of this interface. The point is, roughly, that wishful seeing is avoided by requiring interactions with utilities to occur *after*—not *during*—perceptual integration.) So, again, the moral seems to be that there must be some mechanisms which cross the domains that input systems establish.

For these and other similar reasons, I assume that there must be relatively nondenominational (i.e., domain-*in*specific) psychological systems which operate, inter alia, to exploit the information that input systems provide. Following the tradition, I shall call these "central" systems, and I will assume that it is the operation of these sorts of systems that people have in mind when they talk, pretheoretically, of such mental processes as thought and problem-solving. Central systems may be domain specific in *some* sense—we will consider this when we get to the issues about 'epistemic boundedness'—but at least they aren't domain specific in the way that input systems are. The interesting question about the central systems is whether, being nondenominational, they are also nonmodular in other respects as well. That is, whether the central systems fail to exhibit the galaxy of properties that lead us to think of the input systems as a natural kind—the properties enumerated in Part III.

Briefly, my argument is going to be this: we have seen that much of what is typical of the input systems is more or less directly a product of their informational encapsulation. By contrast, I'll claim that central systems are, in important respects, *un*encapsulated, and that it is primarily for this reason that they are not plausibly viewed as modular. Notice that I am not going to be arguing for a tautology. It is perfectly possible, in point of logic, that a system which is *not* domain specific might nevertheless be encapsulated. Roughly, domain specificity has to do with the range of questions for which a device provides answers (the range of inputs for which it computes analyses); whereas encapsulation has to do with the range of information that the device consults in deciding what answers to provide. A system could thus be domain specific but unencapsulated (it answers a relatively narrow range of questions but in doing so it uses whatever it knows); and a system could be nondenomi-

national but encapsulated (it will give some answer to any question; but it gives its answers off the top of its head—i.e., by reference to less than all the relevant information). If, in short, it is true that only domain-specific systems are encapsulated, then that truth is interesting. Perhaps it goes without saying that I am not about to demonstrate this putative truth. I am, however, about to explore it.

So much for what I'm going to be arguing *for*. Now a little about the strategy of the argument. The fact is that there is practically no direct evidence, pro or con, on the question whether central systems are modular. No doubt it is possible to achieve some gross factoring of "intelligence' into "verbal" versus "mathematical/spatial" capacities; and no doubt there is something to the idea of a corresponding hemispheric specialization. But such dichotomies are *very* gross and may themselves be confounded with the modularity of the input systems—that is to say, they give very little evidence for the existence of domain-specific (to say nothing of modular) systems other than the ones that subserve the functions of perceptual and linguistic analysis.

When you run out of direct evidence, you might just as well try arguing from analogies, and that is what I propose to do. I have been assuming that the typical function of central systems is the fixation of belief (perceptual or otherwise) by nondemonstrative inference. Central systems look at what the input systems deliver, and they look at what is in memory, and they use this information to constrain the computation of 'best hypotheses' about what the world is like. These processes are, of course, largely unconscious, and very little is known about their operation. However, it seems reasonable enough that something can be inferred about them from what we know about *explicit* processes of nondemonstrative inference—viz., from what we know about empirical inference in science. So, here is how I am going to proceed. First, I'll suggest that scientific confirmation—the nondemonstrative fixation of belief in science—is typically unencapsulated. I'll then argue that if, pursuing the analogy, we assume that the central psychological systems are also unencapsulated, we get a picture of those systems that is, anyhow, not radically implausible given such information about them as is currently available.

The nondemonstrative fixation of belief in science has two prop-

erties which, though widely acknowledged, have not (so far as I know) yet been named. I shall name them: confirmation in science is *isotropic* and it is *Quineian*. It is notoriously hard to give anything approaching a rigorous account of what being isotropic and Quineian amounts to, but it is easy enough to convey the intuitions.

By saying that confirmation is isotropic, I mean that the facts relevant to the confirmation of a scientific hypothesis may be drawn from anywhere in the field of previously established empirical (or, of course, demonstrative) truths. Crudely: everything that the scientist knows is, in principle, relevant to determining what else he ought to believe. In principle, our botany constrains our astronomy, if only we could think of ways to make them connect.

As is usual in a methodological inquiry, it is possible to consider the isotropy of confirmation either normatively (as a principle to which we believe that rational inductive practice *ought* to conform) or sociologically (as a principle which working scientists actually adhere to in assessing the degree of confirmation of their theories). In neither case, however, should we view the isotropy of confirmation as merely gratuitous—or, to use a term of Rorty's (1979) as merely "optional." If isotropic confirmation 'partially defines the language game that scientists play' (remember when we used to talk that way?), that is because of a profound conviction—partly metaphysical and partly epistemological—to which scientists implicitly subscribe: the world is a connected causal system *and we don't know how the connections are arranged.* Because we don't, we must be prepared to abandon previous estimates of confirmational relevance as our scientific theories change. The points of all this is: confirmational isotropy is a reasonable property for nondemonstrative inference to have because the goal of nondemonstrative inference is to determine the truth about a causal mechanism—the world—of whose workings we are arbitrarily ignorant. That is why our institution of scientific confirmation is isotropic, and it is why it is plausible to suppose that what psychologists call "problem-solving" (i.e., nondemonstrative inference in the service of individual fixation of belief) is probably isotropic too.

The isotropy of scientific confirmation has sometimes been denied, but never, I think, very convincingly. For example, according to some historians it was part of the Aristotelian strategy against Galileo to claim that no data other than observations of the movements

of astronomical objects could, in principle, be relevant to the (dis)confirmation of the geocentric theory. Telescopic observations of the phases of Venus were thus ruled irrelevant a priori. In notably similar spirit, some linguists have recently claimed that no data except certain specified kinds of facts about the intuitions of native speakers could, in principle, be relevant to the (dis)confirmation of grammatical theories. Experimental observations from psycholinguistics are thus ruled irrelevant a priori. However, this sort of methodology seems a lot like special pleading: you tend to get it precisely when cherished theories are in trouble from prima facie disconfirming data. Moreover, it often comports with Conventionalist construals of the theories so defended. That is, theories for which nonisotropic confirmation is claimed are often viewed, even by their proponents, as merely mechanisms for making predictions; what is alleged in their favor is predictive adequacy rather than correspondence to the world. (Viewed from our perspective, nonisotropic confirmation is, to that extent, not a procedure for fixation of belief, since, on the Conventionalist construal, the predictive adequacy of a theory is *not* a reason for believing that the theory is *true*.)

One final thought on the isotropy issue. We are interested in isotropic systems because such systems are ipso facto unencapsulated. We are interested in scientific confirmation because (a) there is every reason to suppose that it is isotropic; (b) there is every reason to suppose that it is a process fundamentally similar to the fixation of belief; and (c) it is perhaps the only "global", unencapsulated, wholistic cognitive process about which anything is known that's worth reporting. For all that, scientific *confirmation* is probably not the best place to look if you want to see cognitive isotropy writ large. The best place to look, at least if one is willing to trust the anecdotes, is scientific *discovery*.

What the anecdotes say about scientific discovery—and they say it with a considerable show of univocality (see, e.g., papers in Ortony, 1979)—is that some sort of 'analogical reasoning' often plays a central role. It seems to me that we are thoroughly in the dark here, so I don't propose to push this point very hard. But it really does look as though there have been frequent examples in the history of science where the structure of theories in a new subject area has been borrowed from, or at least suggested by,

theories *in situ* in some quite different domain: what's known about the flow of water gets borrowed to model the flow of electricity; what's known about the structure of the solar system gets borrowed to model the structure of the atom; what's known about the behavior of the market gets borrowed to model the process of natural selection, which in turn gets borrowed to model the shaping of operant responses. And so forth. The point about all this is that "analogical reasoning" would seem to be isotropy in the purest form: a process which depends precisely upon the transfer of information among cognitive domains previously assumed to be mutually irrevelant. By definition, encapsulated systems do not reason analogically.

I want to suggest two morals before I leave this point. The first is that the closer we get to what we are pretheoretically inclined to think of as the 'higher,' 'more intelligent', less reflexive, less routine exercises of cognitive capacities, the more such global properties as isotropy tend to show up. I doubt that this is an accident. I suspect that it is precisely its possession of such global properties that we have in mind when we think of a cognitive process as paradigmatically intelligent. The second moral preshadows a point that I shall jump up and down about further on. It is striking that, while everybody thinks that analogical reasoning is an important ingredient in all sorts of cognitive achievements that we prize, nobody knows anything about how it works; not even in the dim, in-a-glass-darkly sort of way in which there are some ideas about how confirmation works. I don't think that this is an accident either. In fact, I should like to propose a generalization; one which I fondly hope will some day come to be known as 'Fodor's First Law of the Nonexistence of Cognitive Science'. It goes like this: the more global (e.g., the more isotropic) a cognitive process is, the less anybody understands it. *Very* global processes, like analogical reasoning, aren't understood at all. More about such matters in the last part of this discussion.

By saying that scientific confirmation is Quineian, I mean that the degree of confirmation assigned to any given hypothesis is sensitive to properties of the entire belief system; as it were, the shape of our whole science bears on the epistemic status of each scientific hypothesis. Notice that being Quineian and being isotropic are not the same properties, though they are intimately related. For example, if scientific confirmation is isotropic, it is quite possible

that some fact about photosynthesis in algae should be relevant to the confirmation of some hypothesis in astrophics ("the universe in a grain of sand" and all that). But the point about being Quineian is that we might have two astrophysical theories, both of which make the same predictions about algae and about everything else that we can think of to test, but such that one of the theories is better confirmed than the other—e.g., on grounds of such considerations as simplicity, plausibility, or conservatism. The point is that simplicity, plausibility, and conservatism are properties that theories have in virtue of their relation to the whole structure of scientific beliefs *taken collectively*. A measure of conservatism or simplicity would be a metric over *global* properties of belief systems.

Consider, by way of a simple example, Goodman's original (1954) treatment of the notion of projectability. We know that two hypotheses that are equivalent in respect of all the available data may nevertheless differ in their level of confirmation depending on which is the more projectable. Now, according to Goodman's treatment, the projectability of a hypothesis is inherited (at least in part) from the projectability of its vocabulary, and the projectability of an item of scientific vocabulary is determined by the (weighted?) frequency with which that item *has been projected* in previously successful scientific theories. So, the whole history of past projections contributes to determining the projectability of any given hypothesis on Goodman's account, and the projectability of a hypothesis (partially) determines its level of confirmation. Similarly with such notions as simplicity, conservatism, and the rest if only we knew how to measure them.

The idea that scientific confirmation is Quineian is by no means untendentious. On the contrary, it was a legacy of traditional philosophy of science—one of the "dogmas of Empiricism" (Quine, 1953) that there must be *semantic* connections between each theory statement and some data statements. That is, each hypothesis about "unobservables" must *entail* some predictions about observables, such entailments holding in virtue of the meanings of the theoretical terms that the hypotheses contain.[38] The effect of postulating such connections would be to determine a priori that certain data would disconfirm certain hypotheses, *whatever the shape of the rest of one's science might be*. For, of course, if H entails O, the discovery that –O would entail that –H. To that extent, the (dis)confirmation of

H by –O is independent of global features of the belief system that H and O belong to. To postulate meaning relations between data statements and theory statements is thus to treat confirmation as a *local* phenomenon rather than a global one.

I emphasize this consideration because analogous semantic proposals can readily be found in the psychological literature. For example, in the sorts of cognitive theories espoused by, say, Bruner or Vygotsky (and, more recently, in the work of the "procedural" semanticists), it is taken for granted that there must be connections of meaning between 'concepts' and 'percepts'. Basically, according to such theories, concepts are recipes for sorting stimuli into categories. Each recipe specifies a (more or less determinate) galaxy of tests that one can perform to effect a sorting, and each stimulus category is identified with a (more or less determinate) set of outcomes of the tests. To put the idea crudely but near enough for present purposes, there's a rule that you can test for *dog* by finding out if a thing barks, and the claim is that this rule is constitutive (though not, of course, exhaustive) of the concept *dog*. Since it is alleged to be a *conceptual* truth that whether it barks is relevant to whether it's a dog, it follows that the confirmation relation between "a thing is a dog" and "it barks" is insensitive to global properties of one's belief system. So considerations of theoretical simplicity etc. *could* not, even in principle, lead to the conclusion that whether it barks is *ir*relevant to whether it's a dog. To embrace that conclusion would be to change the concept.

This sort of example makes it clear how closely related being Quineian and being isotropic are. Since, on the view just scouted, it is a matter of *meaning* that barking is relevant to dogness, it is not possible to discover on empirical grounds that one was wrong about that relevancy relation. But isotropy is the principle that *any* fact may turn out to be (ir)relevant to the confirmation of any other. The Bruner-Vygotsky-procedural semantics line is thus incompatible with the isotropy of confirmation as well as with its Quineianness.

In saying that confirmation is isotropic and Quineian, I am thus consciously disgreeing with major traditions in the philosophy of science and in cognitive psychology. Nevertheless, I shall take it for granted that scientific confirmation is Quineian and isotropic. (Those who wish to see the arguments should refer to such classic papers in the modern philosophy of science as Quine, 1953, and

Putnam, 1962.) Moreover, since I am committed to relying upon the analogy between scientific confirmation and psychological fixation of belief, I shall take it for granted that the latter must be Quineian and isotropic too, hence that the Bruner-Vygotsky-procedural semantics tradition in cognitive psychology must be mistaken. I propose, at this point, to be both explicit and emphatic. The argument is that the central processes which mediate the fixation of belief are typically processes of rational nondemonstrative inference and that, since processes of rational nondemonstrative inference are Quineian and isotropic, so too are central processes. In particular, the theory of such processes must be consonant with the principle that the level of acceptance of any belief is sensitive to the level of acceptance of any other and to global properties of the field of beliefs taken collectively.

Given these assumptions, I have now got two things to do: I need to show that this picture of the central processes is broadly incompatible with the assumption that they are modular, and I need to show that it is a picture that has some plausibility independent of the putative analogy between cognitive psychology and the philosophy of science.

I take it that the first of these claims is relatively uncontroversial. We argued that modularity is fundamentally a matter of informational encapsulation and, of course, informationally encapsulated is precisely what Quineian/isotropic systems are not. When we discussed input systems, we thought of them as mechanisms for projecting and confirming hypotheses. And we remarked that, viewed that way, the informational encapsulation of such systems is tantamount to a constraint on the confirmation metrics that they employ; the confirmation metric of an encapsulated system is allowed to 'look at' only a certain restricted class of data in determining which hypothesis to accept. If, in particular, the flow of information through such a system is literally bottom-to-top, then its informational encapsulation consists in the fact that the ith-level hypotheses are (dis)confirmed solely by reference to lower-than-ith level representations. And even if the flow of data is unconstrained *within* a module, encapsulation implies constraints upon the access of intramodular processes to extramodular information sources. Whereas, by contrast, isotropy is by definition the property that a system has when it can look at anything it knows about in the

course of determining the confirmation levels of hypotheses. So, in general, the more isotropic a confirmation metric is, the more heterogeneous the provenance of the data that it accepts as relevant to constraining its decisions. Scientific confirmation is isotropic in the limit in this respect; it provides a model of what the *non*modular fixation of belief is like.

Similarly with being Quineian. Quineian confirmation metrics are ipso facto sensitive to global properties of belief systems. Now, an informationally encapsulated system *could*, strictly speaking, nevertheless be Quineian. Simplicity, for example, could constrain confirmation even in a system which computes its simplicity scores over some arbitrarily selected subset of beliefs. But this is mere niggling about the letter. In spirit, global criteria for the evaluation of hypotheses comport most naturally with isotropic principles for the relevance of evidence. Indeed, it is only on the assumption that the selection of evidence is isotropic that considerations of simplicity (and other such global properties of hypotheses) are *rational* determinants of belief. It is epistemically interesting that H & T is a simpler theory than -H & T where H is a hypothesis to be evaluated and T is the rest of what one believes. But there is no interest in the analogous consideration where T is some *arbitrarily delimited* subset of one's beliefs. Where relevance is non-isotropic, assessments of relative simplicity can be gerrymandered to favor any hypothesis one likes. This is one of the reasons why the operation of (by assumption informationally encapsulated) input systems should not be identified with the fixation of perceptual belief; not, at least, by those who wish to view the fixation of perceptual belief as by and large a rational process.

So it seems clear that isotropic/Quineian systems are ipso facto unencapsulated; and if unencapsulated, then presumably non-modular. Or rather, since this is all a matter of degree, we had best say that *to the extent that* a system is Quineian and isotropic, it is also nonmodular. If, in short, isotropic and Quineian considerations are especially pressing in determining the course of the computations that central systems perform, it should follow that these systems differ in their computational character from the vertical faculties.

We are coming close to what we started out to find: an overall taxonomy of cognitive systems. According to the present proposal,

there are, at a minimum, two families of such systems: modules (which are, relatively, domain specific and encapsulated) and central processes (which are, relatively, domain neutral and isotropic/Quinean). We have suggested that the characteristic function of modular cognitive systems is input analysis and that the characteristic function of central processes is the fixation of belief. If this is right, then we have three ways of taxonomizing cognitive processes which prove to be coextensive:

FUNCTIONAL TAXONOMY: input analysis versus fixation of belief
TAXONOMY BY SUBJECT MATTER: domain specific versus domain neutral
TAXONOMY BY COMPUTATIONAL CHARACTER: encapsulated versus Quinean/isotropic

I repeat that this coextension, if it holds at all, holds contingently. Nothing in point of logic stops one from imagining that these categories cross-classify the cognitive systems. If they do not, then that is a fact about the structure of the mind. Indeed, it is a *deep* fact about the structure of the mind.

All of which would be considerably more impressive if there were better evidence for the view of central processes that I have been proposing. Thus far, that account rests entirely on the analogy between psychological processes of belief fixation and a certain story about the character of scientific confirmation. There is very little that I can do about this, given the current underdeveloped state of psychological theories of thought and problem-solving. For what it's worth, however, I want to suggest two considerations that seem relevant and promising.

The first is that the difficulties we encounter when we try to construct theories of central processes are just the sort we would expect to encounter if such processes are, in essential respects, Quinean/isotropic rather than encapsulated. The crux in the construction of such theories is that there seems to be no way to delimit the sorts of informational resources which may affect, or be affected by, central processes of problem-solving. We can't, that is to say, plausibly view the fixation of belief as effected by computations over bounded, local information structures. A graphic example of this sort of difficulty arises in AI, where it has come to be known as the "frame problem" (i.e., the problem of putting a "frame"

around the set of beliefs that may need to be revised in light of specified newly available information. Cf. the discussion in McCarthy and Hayes, 1969, from which the following example is drawn).

To see what's going on, suppose you were interested in constructing a robot capable of coping with routine tasks in familiar human environments. In particular, the robot is presented with the job of phoning Mary and finding out whether she will be late for dinner. Let's assume that the robot 'knows' it can get Mary's number by consulting the directory. So it looks up Mary's number and proceeds to dial. So far, so good. But now, notice that commencing to dial has all sorts of direct and indirect effects on the state of the world (including, of course, the internal state of the robot), and some of these effects are ones that the device needs to keep in mind for the guidance of its future actions and expectations. For example, when the dialing commences, the phone ceases to be free to outside calls; the robot's fingers (or whatever) undergo appropriate alterations of spatial location; the dial tone cuts off and gets replaced by beeps; something happens in a computer at Murray Hill; and so forth. Some (but, in principle, not all) such consequences are ones that the robot must be designed to monitor since they are relevant to "updating" beliefs upon which it may eventually come to act. Well, *which* consequences? The problem has at least the following components. The robot must be able to identify, with reasonable accuracy, those of its previous beliefs whose truth values may be expected to alter as a result of its current activities; and it must have access to systems that do whatever computing is involved in effecting the alterations.

Notice that, unless these circuits are arranged correctly, things can go absurdly wrong. Suppose that, having consulted the directory, the robot has determined that Mary's number is 222-2222, which number it commences to dial, pursuant to instructions previously received. But now it occurs to the machine that *one of the beliefs that may need updating in consequence of its having commenced dialing is its (recently acquired) belief about Mary's telephone number.* So, of course, it stops dialing and goes and looks up Mary's telephone number (again). Repeat, *da capo*, as many times as may amuse you. Clearly, we have here all the makings of a computational trap. Unless the robot can be assured that some of its beliefs are invariant under some of its actions, it will never get to *do* anything.

How, then, does the machine's program determine which beliefs the robot ought to reevaluate given that it has embarked upon some or other course of action? What makes this problem so hard is precisely that it seems unlikely that any *local* solution will do the job. For example, the following truths appear to be self-evident: First, that there is no fixed set of beliefs such that, for any action, those and only those beliefs are the ones that require reconsideration. (That is, which beliefs are up for grabs depends intimately upon which actions are performed and upon the context of the performances. There are *some*—indeed, indefinitely many—actions which, if performed, *should* lead one to consider the possibility that Mary's telephone number has changed in consequence.) Second, new beliefs don't come docketed with information about which old beliefs they ought to affect. On the contrary, we are forever being surprised by the implications of what we know, including, of course, what we know about the actions we perform. Third, the set of beliefs apt for reconsideration cannot be determined by reference to the recency of their acquisition, or by reference to their generality, or by reference to merely semantic relations between the contents of the beliefs and the description under which the action is performed . . . etc. Should any of these propositions seem *less* than self-evident, consider the special case of the frame problem where the robot is a mechanical scientist and the action performed is an experiment. Here the question 'which of my beliefs ought I to reconsider given the possible consequences of my action' is transparently equivalent to the question "What, in general, is the optimal adjustment of my beliefs to my experiences?" This is, of course, exactly the question that a theory of confirmation is supposed to answer; and, as we have been at pains to notice, confirmation is not a relation reconstructible by reference to local properties of hypotheses or of the data that bear upon them.

I am suggesting that, as soon as we begin to look at cognitive processes other than input analysis—in particular, at central processes of nondemonstrative fixation of belief—we run into problems that have a quite characteristic property. They seem to involve isotropic and Quinean computations; computations that are, in one or other respect, sensitive to the whole belief system. This is exactly what one would expect on the assumption that nondemonstrative fixation of belief really is quite like scientific confirmation, and that

scientific confirmation is itself characteristically Quineian and isotropic. In this respect, it seems to me, the frame problem is paradigmatic, and in this respect the seriousness of the frame problem has not been adequately appreciated.

For example, Raphael (1971) comments as follows: "(An intelligent robot) will have to be able to carry out tasks. Since a task generally involves some change in the world, it must be able to update its model (of the world) so it remains as accurate during and after the performance of a task as it was before. Moreover, it must be able to *plan* how to carry out a task, and this planning process usually requires keeping 'in mind' simultaneously a variety of possible actions and corresponding models of hypothetical worlds that would result from those actions. The bookkeeping problems involved with keeping track of these hypothetical worlds account for much of the difficulty of the frame problem" (p. 159). This makes it look as though the problem is primarily (a) how to notate the possible worlds and (b) how to keep track of the *demonstrative* consequences of changing state descriptions. But the deeper problem, surely, is to keep track of the *non*demonstrative consequences. Slightly more precisely, the problem is, given an arbitrary belief world W and a new state description 'a is F', what is the appropriate successor belief world W'? What ought the device to believe, given that it used to believe W and now believes that a is F? But this isn't just a bookkeeping problem; it is the general problem of inductive confirmation.[39]

So far as I can tell, the usual assumption about the frame problem in AI is that it is somehow to be solved 'heuristically'. The idea is that, while nondemonstrative confirmation (and hence, presumably, the psychology of belief fixation) is isotropic and Quineian *in principle*, still, given a particular hypothesis, there are, in practice, heuristic procedures for determining the range of effects its acceptance can have on the rest of one's beliefs. Since these procedures are by assumption merely heuristic, they may be assumed to be local—i.e., to be sensitive to less than the whole of the belief systems to which they apply. Something like this may indeed be true; there is certainly considerable evidence for heuristic short-cutting in belief fixation, deriving both from studies of the psychology of problem-solving (for a recent review, see Nisbett and Ross, 1980) and from the sociology of science (Kuhn, 1970). In such cases, it is possible

to show how potentially relevant considerations are often systematically ignored, or distorted, or misconstrued in favor of relatively local (and, of course, highly fallible) problem-solving strategies. Perhaps a bundle of such heuristics, properly coordinated and rapidly deployed, would suffice to make the central processes of a robot as Quineian and isotropic as yours, or mine, or the practicing scientist's ever actualy succeed in being. Since there are, at present, no serious proposals about what heuristics might belong to such a bundle, it seems hardly worth arguing the point.

Still, I am going to argue it a little.

There are those who hold that ideas recently evolved in AI—such notion as, e.g., those of 'frame' (see Minsky, 1975)[40] or 'script' (see Schank and Abelson, 1975)—will illuminate the problems about the globality of belief fixation since they do, in a certain sense, provide for placing a frame around the body of information that gets called when a given sort of problem is encountered. (For a discussion that runs along these optimistic lines, see Thagard, 1980.) It seems to me, however, that the appearance of progress here is entirely illusory—a prime case of confusing a notation with a theory.

If there were a principled solution to the frame problem, then no doubt that solution could be expressed as a constraint on the scripts, or frames, to which a given process of induction has access. But, lacking such a solution, there is simply no content to the idea that only the information represented in the frame (/script) that a problem elicits is computationally available for solving the problem. For one thing, since there are precisely no constraints on the individuation of frames (/scripts), *any* two pieces of information can belong to the same frame (/script) at the discretion of the programmer. This is just a way of saying that the solution of the frame problem can be accommodated to the frame (/script) notation *whatever that solution turns out to be.* Which is just another way of saying that the notation does not constrain the solution. Second, it is a widely advertised property of frames (/scripts) that they can cross-reference to one another. The frame for Socrates says, among other things, 'see Plato' . . . and so forth. There is no reason to doubt that, in any developed model, the system of cross-referencing would imply a graph in which there is a route (of greater or lesser length) from each point to any other. But now we have the frame problem all over again, in the form: Which such paths should

actually be traversed in a given case of problem-solving, and what should bound the length of the trip? All that has happened is that, instead of thinking of the frame problem as an issue in the logic of confirmation, we are now invited to think of it as an issue in the theory of executive control (a change which there is, by the way, no reason to assume is for the better). More of this presently.

For now, let's summarize the major line of argument. If we assume that central processes are Quineian and isotropic, then we ought to predict that certain kinds of problems will emerge when we try to construct psychological theories which simulate such processes or otherwise explain them; specifically, we should predict problems that involve the characterization of nonlocal computational mechanisms. By contrast, such problems should not loom large for theories of psychological modules. Since, by assumption, modular systems are informationally encapsulated, it follows that the computations they perform are relatively local. It seems to me that these predictions are in reasonably good accord with the way that the problems of cognitive science have in fact matured: the input systems appear to be primarily stimulus driven, hence to exploit computational processes that are relatively insensitive to the general structure of the organism's belief system. Whereas, when we turn to the fixation of belief, we get a complex of problems that appear to be intractable precisely because they concern mental processes that aren't local. Of these, the frame problem is, as we have seen, a microcosm.

I have been marshaling considerations in favor of the view that central processes are Quineian/isotropic. That is what the analogy to scientific confirmation suggests that they ought to be, and the structure of the problems that arise in attempts to model central processes is quite compatible with that view of them. I now add that the view of central processes as computationally global can perhaps claim some degree of neurological plausibility. The picture of the brain that it suggests is a reasonably decent first approximation to the kind of brain that it appears we actually have.

When we discussed input analyzers, I commented on the natural connection between informational encapsulation and fixed neural architecture. Roughly, standing restrictions on information flow imply the option of hardwiring. If, in the extreme case, system B is required to take note of information from system A and is allowed

to take note of information from nowhere else, you might as well build your brain with a permanent neuroanatomical connection from A to B. It is, in short, reasonable to expect biases in the distribution of information to mental processes to show up as structural biases in neural architecture.

Consider, by contrast, Quineian/isotropic systems, where more or less any subsystem may want to talk to any other at more or less any time. In this case, you'd expect the corresponding neuroanatomy to be relatively diffuse. At the limit, you might as well have a random net, with each computational subsystem connected, directly or indirectly, with every other; a kind of wiring in which you get a minimum of stable correspondence between neuroanatomical form and psychological function. The point is that in Quineian/isotropic systems, it may be *unstable, instantaneous* connectivity that counts. Instead of hardwiring, you get a connectivity that changes from moment to moment as dictated by the interaction between the program that is being executed and the structure of the task in hand. The moral would seem to be that computational isotropy comports naturally with neural isotropy (with what Lashley called "equipotentiality" of neural structure) in much the same way that informational encapsulation comports naturally with the elaboration of neural hardwiring.

So, if input analysis is modular and thought is Quineian/isotropic, you might expect a kind of brain in which there is stable neural architecture associated with perception-and-language but not with thought. And, I suggest, this seems to be pretty much what we in fact find. There is, as I remarked above, quite a lot that can be said about the neural specificity of the perceptual and linguistic mechanisms: at worst we can enumerate in some detail the parts of the brain that handle them; and at best we can exhibit characteristic neural architecture in the areas where these functions are performed. And then there are the rest of the higher brain systems (cf. what used to be called "association cortex"), in which neural connectivity appears to go every which way and the form/function correspondence appears to be minimal. There is some historical irony in all this. Gall argued from a (vertical) faculty psychology to the macroscopic differentiation of the brain. Flourens, his archantagonist, argued from the unity of the Cartesian ego to the brain's equipotentiality (see Bynum, op. cit.). The present suggestion is that they were *both* right.[41]

I am, heaven knows, not about to set up as an expert on neuropsychology, and I am painfully aware how impressionistic this all is. But while we're collecting impressions, I think the following one is striking. A recent issue of *Scientific American* (September, 1979) was devoted to the brain. Its table of contents is quite as interesting as the papers it contains. There are, as you might expect, articles that cover the neuropsychology of language and of the perceptual mechanisms. But there is nothing on the neuropsychology of thought—presumably because nothing is known about the neuropsychology of thought. I am suggesting that there is a good reason why nothing is known about it—namely, that there is nothing *to* know about it. You get form/function correspondence for the modular processes (specifically, for the input systems); but, in the case of central processes, you get an approximation to universal connectivity, hence no stable neural architecture to write *Scientific American* articles about.

To put these claims in a nutshell; there are *no* content-specific central processes for the performance of which correspondingly specific neural structures have been identified. Everything we now know is compatible with the claim that central problem-solving is subserved by equipotential neural mechanisms. This is precisely what you would expect if you assume that the central cognitive processes are largely Quineian and isotropic.

PART V
CAVEATS AND CONCLUSIONS

We now have before us what might be called a 'modified' modularity theory of cognitive processes. According to this theory, Gall was right in claiming that there are vertical faculties (domain specific computational mechanisms). Indeed, a still stronger claim is plausible: that the vertical faculties are modules (informationally encapsulated, neurologically hardwired, innately specified and so forth). But nonmodular cognitive systems are also acknowledged, and it is left open that these latter may exhibit features of horizontal organization. Roughly speaking, on this account, the distinction between vertical and horizontal modes of computational organization is taken to be coextensive with the functional distinction